

OMSK, Russian Federation

WMO: 286980

Lat: 55.017N

Lon: 73.383E

Elev: 401

StdP: 14.48

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.5	(c) -22.9	(d) -34.0	(e) 0.8	(f) -27.6	(g) -29.1	(h) 1.1	(i) -22.8	(j) 24.0	(k) 21.4	(l) 21.2	(m) 17.7	(n) 4.3	(o) 60	(p) 0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	87.8	65.7	84.2	64.9	80.9	63.7	69.2	81.7	67.5	79.5	66.0	77.1	9.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 64.8	(b) 93.1	(c) 73.2	(d) 63.0	(e) 87.5	(f) 71.9	(g) 61.7	(h) 83.5	(i) 71.0	(j) 33.5	(k) 81.7	(l) 32.2	(m) 79.7	(n) 31.0	(o) 77.3	(p) 76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.8	17.4	-32.6	93.0	5.7	3.7	-36.7	95.6	-40.0	97.7	-43.3	99.8	-47.4	102.4
			-32.6	72.3	5.7	1.8	-36.6	73.6	-40.0	74.7	-43.1	75.7	-47.3	77.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	0.1	5.1	19.9	40.1	54.5	64.1	67.0	62.8	51.5	38.5	20.0	7.0		
	DBStd	25.85	13.70	12.53	10.57	10.79	9.67	7.95	6.61	7.37	8.27	8.90	13.65	13.83		
	HDD50	6814	1547	1258	932	328	64	5	0	2	78	367	900	1333		
	HDD65	10858	2012	1678	1397	746	345	111	59	132	410	820	1350	1798		
	CDD50	1726	0	0	0	32	205	429	526	399	124	11	0	0		
	CDD65	296	0	0	0	0	21	85	120	63	7	0	0	0		
	CDH74	3104	0	0	0	13	348	899	1116	622	106	0	0	0		
	CDH80	921	0	0	0	1	86	279	340	193	22	0	0	0		
	WSAvg	7.8	7.1	7.2	8.6	9.1	8.9	7.5	7.3	7.1	7.4	7.9	8.2	7.8		
	PrecAvg	15.8	0.8	0.6	0.6	0.9	1.3	2.1	2.6	2.2	1.2	1.3	1.3	1.0		
	PrecMax	25.0	1.9	1.7	1.9	2.5	4.2	5.4	7.1	5.7	3.5	3.5	3.3	3.7		
(15) Precipitation	PrecMin	10.2	0.1	0.0	0.1	0.1	0.2	0.2	0.0	0.5	0.1	0.2	0.1	0.0		
	PrecStd	3.2	0.4	0.3	0.4	0.6	0.8	1.2	1.5	1.1	0.7	0.7	0.6	0.6		
	DB	33.4	34.2	45.3	75.3	87.3	91.0	92.9	91.7	82.8	67.9	46.2	34.9	33.2		
	MCWB	32.4	32.4	39.1	55.9	61.8	66.6	68.7	67.2	59.9	52.3	42.9	33.0	30.0		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	28.7	31.4	39.5	68.1	82.1	86.4	87.4	84.6	76.7	62.3	42.4	32.4	28.4		
	2% MCWB	27.5	29.8	35.3	52.0	60.0	64.9	67.2	65.7	58.3	50.0	40.0	31.3	27.4		
	5% DB	25.1	28.1	36.3	63.1	77.8	83.1	83.9	80.7	70.5	57.0	39.1	29.8	25.8		
	5% MCWB	23.9	26.7	33.5	50.1	58.1	64.4	66.1	64.5	56.3	47.5	36.7	28.6	24.8		
	10% DB	21.1	23.9	34.1	57.5	73.1	79.2	80.5	76.7	65.9	52.0	35.4	25.7	21.7		
	10% MCWB	20.1	22.6	31.8	47.3	55.9	62.9	65.5	63.1	54.0	45.1	33.5	24.4	20.7		
	0.4% WB	31.9	32.8	39.4	57.5	65.4	70.7	71.7	70.0	62.9	54.7	43.3	33.2	29.7		
	0.4% MCDB	32.7	33.9	43.7	70.7	82.3	84.4	84.4	83.6	77.2	64.1	44.8	34.0	29.7		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	27.5	29.6	35.8	53.8	61.8	68.1	69.7	67.8	59.7	51.3	39.8	31.3	27.4		
	2% MCDB	28.6	30.9	38.4	64.9	77.7	81.1	81.7	80.0	71.8	59.4	41.9	32.4	28.4		
	5% WB	24.0	26.4	33.9	50.9	59.3	66.2	68.1	66.1	57.4	48.6	36.5	28.3	24.8		
	5% MCDB	25.1	27.7	36.0	62.1	74.2	78.4	80.1	76.9	68.9	55.6	38.5	29.5	25.8		
	10% WB	20.1	22.5	31.9	47.8	57.1	64.4	66.6	64.3	54.9	45.6	33.4	24.4	21.7		
	10% MCDB	21.1	23.8	34.1	57.0	70.6	76.2	77.7	73.9	64.8	51.0	34.8	25.5	22.7		
	MDBR	15.0	16.4	16.3	18.5	22.8	20.7	19.4	19.7	19.2	15.2	11.9	14.0	12.0		
	5% DB	16.0	13.8	14.4	27.3	29.6	26.3	25.3	26.6	27.6	24.3	12.1	13.9	12.0		
(36) Mean Daily Temperature Range	MCWBR	15.8	13.1	11.5	15.4	14.6	11.3	10.2	11.6	13.1	14.4	10.5	13.8	12.0		
	MCDBR	16.0	13.2	13.1	25.6	27.1	23.0	22.0	22.4	25.3	21.7	11.8	14.1	12.0		
	MCWBR	15.9	12.7	11.1	15.4	15.1	11.1	10.2	11.0	12.9	13.8	10.6	14.1	12.0		
	MCDBR	15.9	12.7	11.1	15.4	15.1	11.1	10.2	11.0	12.9	13.8	10.6	14.1	12.0		
(40) Clear-Sky Solar Irradiance	taub	0.284	0.316	0.375	0.406	0.402	0.409	0.434	0.432	0.370	0.354	0.323	0.289	0.289		
	taud	2.044	2.015	1.931	2.128	2.229	2.289	2.214	2.186	2.323	2.237	2.101	2.007	2.007		
	Ebn at Noon	193	231	243	257	266	266	255	247	249	219	175	157	157		
	Edh at Noon	22	32	45	43	41	39	41	40	30	26	21	18	18		
(44) All-Sky Solar Radiation	RadAvg	309	652	1148	1474	1781	1900	1748	1410	997	517	250	199	199		
	RadStd	22	30	90	117	152	113	126	112	81	63	36	23	23		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page